MAT 3007 Homework # 3 Spring 2024

Due: Wednesday, March 6 at 12:00pm (noon)

Please note: You should submit your answers (scanned or typeset) in a PDF file via Blackboard. The problem set is due at 12:00pm (noon) on Wednesday, March 6th.

1. (20 pts) Consider the following LP:

$$\begin{aligned} \max \quad & 500x_1 + 250x_2 + 600x_3 \\ \text{s.t.} \quad & 2x_1 + x_2 + x_3 \leq 240 \\ & 3x_1 + x_2 + 2x_3 \leq 150 \\ & x_1 + 2x_2 + 4x_3 \leq 180 \\ & x_1, x_2, x_3 \geq 0 \end{aligned}$$

Use the Simplex method to solve it. For each step, clearly mark what is the current basis, the current basic solution, and the corresponding objective value.

2. (20 pts) Use the Simplex method to solve the following LP. Notice that there is degeneracy and you should select the entering and exiting basis carefully.

$$\begin{aligned} \min \quad & -2x_1 - 3x_2 + x_3 + 12x_4 \\ \text{s.t.} \quad & -2x_1 - 9x_2 + x_3 + 9x_4 + x_5 = 0 \\ & \frac{1}{3}x_1 + x_2 - \frac{1}{3}x_3 - 2x_4 + x_6 = 0 \\ & x_1, x_2, x_3, x_4, x_5, x_6 \geq 0 \end{aligned}$$

3. (20 pts) Use the two-phase method to solve completely the following problem:

$$\begin{array}{lrrrrrcl} \min & 2x_1 & +3x_2 & +3x_3 & +x_4 & -2x_5 & & \\ \text{s.t.} & x_1 & +3x_2 & & +4x_4 & +x_5 & = & 2 \\ & x_1 & +2x_2 & & -3x_4 & +x_5 & = & 2 \\ & -x_1 & -4x_2 & +3x_3 & & & = & 1 \\ & x_1 & , x_2 & , x_3 & , x_4 & , x_5 & \geq & 0 \end{array}$$

4. (20 pts) While solving a standard form LP, we arrive at the following simplex tableau (Table 1) with basic variables x_3, x_4, x_5.

B	δ	-2	0	0	0	-10
3	-1	η	1	0	0	4
4	α	-4	0	1	0	1
5	γ	3	0	0	1	β

Table 1: Simplex tableau for a standard form LP

The entries α, β, γ, δ and η in the tableau are unknown parameters. For each one of the following statements, find the sufficient conditions of the parameter values that will make the statement true.

(a) The LP is unbounded, i.e., the optimal value is $-\infty$.

(b) The current solution is feasible but not optimal.

(c) The current solution has the optimal objective value and there are multiple set of basis that achieve the same objective value.

5. (20 pts) Consider a linear optimization problem in the standard form, described in terms of the following initial tableau (Table 2):

B	0	0	0	δ	3	γ	ξ	0
2	0	1	0	α	1	0	3	β
3	0	0	1	-2	2	η	-1	2
1	1	0	0	0	-1	2	1	3

Table 2: Initial tableau

The entries α, β, γ, δ, η and ξ in the tableau are unknown parameters, and $B = \{2, 3, 1\}$. For each of the following statements, find the sufficient conditions of the parameter values that will make the statement true.

(a) This is an acceptable initial tableau, i.e., the basic variables are feasible for the problem.

(b) The first row (in the constraint) indicates that the problem is infeasible.

(c) The basic solution is feasible but we have not reached an optimal basic set B.

(d) The basic solution is feasible and the first simplex iteration indicates that the problem is unbounded.

(e) The basic solution is feasible, x_6 is a candidate for entering B, and when we choose x_6 as the entering basis, x_3 leaves B.